

Step Detection for Rollator Users with Smartwatches

Denys J.C. Matthies^{1,2}, Marian Haescher¹, Suranga Nanayakkara², Gerald Bieber¹

¹Fraunhofer Institute for Computer Graphics Research IGD Rostock, Rostock, Germany

²Augmented Human Lab, Auckland Bioengineering Institute, The University of Auckland, Auckland, New Zealand
{firstname}@ahlab.org, {firstname.surname}@igd-r.fraunhofer.de

ABSTRACT

Smartwatches enable spatial user input, namely for the continuous tracking of physical activity and relevant health parameters. Additionally, smartwatches are experiencing greater social acceptability, even among the elderly. While step counting is an essential parameter to calculate the user's spatial activity, current detection algorithms are insufficient for calculating steps when using a rollator, which is a common walking aid for elderly people. Through a pilot study conducted with eight different wrist-worn smart devices, an overall recognition of ~10% was achieved. This is because characteristic motions utilized by step counting algorithms are poorly reflected at the user's wrist when pushing a rollator. This issue is also present among other spatial activities such as pushing a pram, a bike, and a shopping cart. This paper thus introduces an improved step counting algorithm for wrist-worn accelerometers. This new algorithm was first evaluated through a controlled study and achieved promising results with an overall recognition of ~85%. As a follow-up, a preliminary field study with randomly selected elderly people who used rollators resulted in similar detection rates of ~83%. To conclude, this research will expectantly contribute to greater step counting precision in smart wearable technology.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility technologies**; Ubiquitous and mobile devices.

KEYWORDS

Smartwatch; Spatial User Input; Step Detection; Rollator; Wheeled Walking Frame; Step Counting; Elderly People.

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1 INTRODUCTION

Assessing spatial activity, such as the user's daily steps is an important parameter. This may indicate potential health conditions and diseases [20], which can help in improving the

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Figure 1. We present a step detection algorithm for elderly people using a rollator, based on a wrist-worn accelerometer found in smartwatches, such as the SimValley AW 429 RX.

dosage of medication. Such tracking can be accomplished with a variety of spatial user interfaces. Smartwatches, in particular, are gaining popularity for several reasons. These devices are becoming more socially acceptable and have increased technological power, with higher sensing capabilities and battery life [27]. Moreover, a wrist-located sensing device has major benefits, being always available at the user's body, while enabling a noninvasive data collection method [19]. An intelligent health monitoring system using an ordinary smartwatch, which tracks vital data, activities, and steps taken [6][30], is highly relevant for the elderly. Those suffering from fatigue when their daily medication fails to coincide with their actual activity performance, will particularly experience a benefit. By acknowledging the benefits of these emerging and powerful wearables, several companies have recently deployed smartwatches among elderly customers [1][3][16][22][24][25], thereby creating an increased social acceptability within this user group. Prior research was conducted in health analysis and step detection with wrist-worn accelerometers [32], including the typical arm movements when walking [9][28]. However, recent studies with Dementia patients [6] show that extracting relevant information only from smartwatches is not trivial and faces technical issues. An emerging problem is that smartwatches have low step detection performance, especially when heavy devices, such as carts or suitcases, are pushed or carried. This problem is particularly significant when the elderly uses a rollator. This is because elderly's general activity level and walking style significantly deviates from younger and healthier users. As the rollator is a common assistive device used by the elderly, developing a reliable step detection for rollator use is highly important. While solely mounting sensors to the walking frame, such as a depth camera [18], force sensor [4], or an IMU [5] appears sufficient, this approach offers limited user tracking.

In this paper, the following contributions are:

- a brief evaluation of current step detections on state-of-the-art wearables to motivate our work,
- an improved step detection algorithm, which is based on a three axis-accelerometer, and
- an evaluation in a lab environment and a field study with elderly people, which both yielded promising accuracy.

2 RELATED WORK

Step detections are usually based on recognizing characteristic reoccurring motions, such as arm swinging [9]. This is commonly measured by an accelerometer, where constant swings over time represents steps [29]. As motion spreads on multiple axes, common step detection algorithms normally select a single axis or calculate a 3d motion vector to count the swings. The most recent commercially available acceleration sensors already include step counters onboard. However, any other cyclic motion can easily generate false positives. Moreover, accurately recognizing low impact steps is difficult, particularly when elderly's use a walking frame [8]. In fact, Clarke et al. [8] investigated this issue at four measurement positions (ankle, thigh, wrist, and waist). While different sensor positions [13] at the body can have a considerable influence on step recognition, speed levels have a significant effect on the results [11]. Generally, common accelerometers used for step detection experience these inaccuracies [12]. Ballesteros et al. [4] and Martins et al. [18], investigated the impact of applying different types of sensors directly to a walking frame to enable step detection. However, a wrist-worn device approach is favorable, since it allows a continuous daily sensing of all spatial activity and vital parameters.

An abundance of research has focused on evaluating and improving step detection without walking aids. Marschollek et al. [17] study particularly focuses on analyzing acceleration sensor signals, while evaluating different algorithms for detecting healthy or impaired walking. Gijbels et al. [14] and Kurtzke et al. [15] explain the effects of physical impairments and the resulting need of walking aids. Their study presents insights into suitable measurement positions and also demonstrates other appropriate application scenarios for step detection, such as a medical context. For instance, a Multiple Sclerosis patients' Expanded Disability Status Scale (EDSS) is capable of being measured by considering the number of steps taken. Although a simple step detection may not be entirely sufficient in determining the correct health status of a Multiple Sclerosis patient, as Schlesinger et al. points out [26], it is still beneficial in determining the disease's stage [23][31]. A diagnostic inclusion of step counting for patients, who required a walking aid after being diagnosed with Multiple Sclerosis, was performed by Adonis et al. [1]. Dijkstra et al. [10] enabled physicians to monitor Parkinson patients and track the diseases' progress, via an accelerometer-based step counting, including step duration and step length.

In summary, the research on step detection mainly focuses on critical applications, such as determining the health status of elderly patients. Therefore, a precise step detection is of great importance. However, literature on rollator use report that

previous step detections are inaccurate, except when the IMU is mounted on the ankle [7]. Whether poor performances still occur with current wrist-mounted wearables, requires verification.

3 PILOTING THE STATE-OF-THE-ART

The actual algorithm of current step detections deployed with state-of-the-art wearables remain unknown. These algorithms, however, are evaluable based on the displayed outcome. Nevertheless, inaccuracies with rollator use is still suspected to occur. To evidence this, a pilot study was conducted with eight consumer wearables on a healthy subject (30yrs, 183cm, 83kg), while using a rollator. With each device (see Figure 2) the male subject walked twice for ~150 steps.



Figure 2. Eight wrist-worn wearables were tested for step counting, which were: No1 D5, No1 D5+, LG Round R, Sony Smartwatch 3, Samsung Gear S, Fitbit Charge 2, Apple Watch 1, and the Garmin VivoFit 3 Wristband.

Although the results (see Figure 3) are only based on two trials with a single subject, substantial differences in recognition rates are not expected to occur with broader sample sizes.

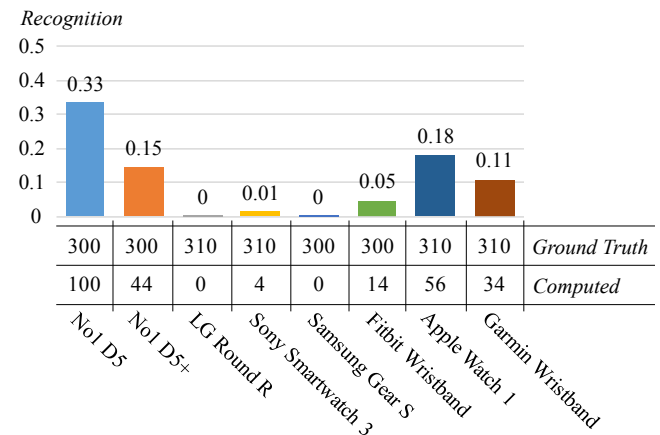


Figure 3. With an average recognition (recall rate) of $M=10.35\%$ $SD=11.5\%$ all devices apparently underperformed due to the marginal wrist motion.

The results revealed a ~10% recognition rate of the tested devices, as they failed to detect steps when using a rollator. Using such erroneous data for health services would result in crucial consequences. This suggests that a new mechanism is needed to track steps, while using devices such as rollators.

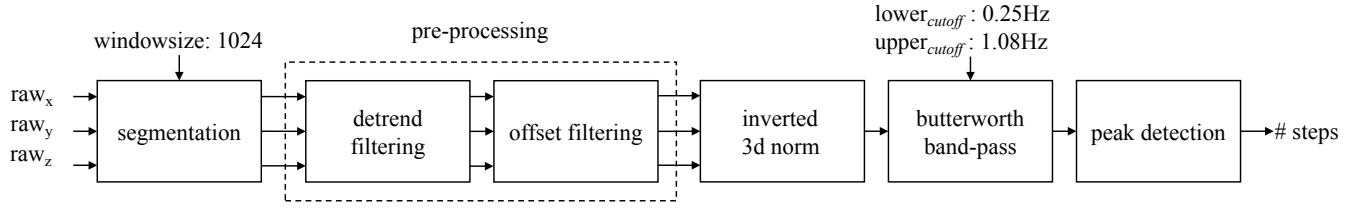


Figure 4. Overview scheme, showing our step detection algorithm, which in particular features an axis fusion.

4 IMPROVED STEP DETECTION ALGORITHM

4.1 Rationale Principle

Each set of steps consists of a back and forth pendulum swing of the arm. The strength of these swings depends on the subject's step impact, which is usually much higher for a healthier and younger subject than for an elderly subject. Here, existing step detection algorithms already reach their limits, in particular when the elderly person uses a rollator. These challenging low-amplitude motions also occur when heavy devices, such as a cart or bag, are pushed or carried. Although arm swinging is now eliminated, miniscule cyclical motions are still present, which results from the upper body being slightly rotated to the left and right while walking. The upper body's twisting motion also transfers to the wrist-mounted sensor, which is useful for step detection. This enables us to design an algorithm for elderlies with a very low step impact, such as those with walking disabilities or those requiring the use of a rollator.

4.2 Implementation

Our accelerometer-based step detection algorithm follows a straight-forward activity recognition approach. [Figure 4](#) shows an overview of all processing steps. Since the algorithm consists of a three axes accelerometer (Bosch BMC050 quantifying at 12bit, sampling at 50Hz in a SimValley AW 429 RX smartwatch – see [Figure 1](#)), raw data was collected in three arrays, while segmenting the data stream into non-overlapping windows (each 1024 values). Subsequently, the data is pre-processed by a detrend and offset-filtering. Following this, the axes are fused into a three-dimensional norm, while it is computed as absolute of each vector. While computing a 3d-norm, a cancellation effect of the signal at these rather low-amplitude motions occurred (see [Figure 5](#)). To overcome this, a helper algorithm was developed to determine

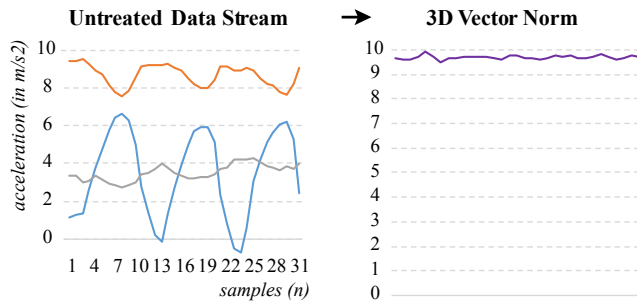


Figure 5. A commonly computed 3d-norm based on untreated xyz-data results in a total cancelation of low-amplitude motion

which of the three axes needed to be inverted. Basically, three 3d-norms were calculated, each with a different inverted axis. The resulting 3d-norm with the highest peak-to-peak delta was then selected for further processing. This selection process is similar to Mortazavi et al. [21] approach. The result is illustrated in [Figure 6](#). By adopting this approach, the algorithm works on both, the left and right arm, while allowing for individual hand positions.

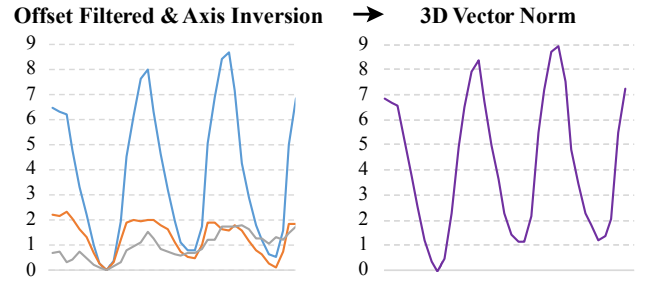


Figure 6. Before computing the 3d-norm, a detrend and offset filtering builds the basis. Then, most essential, a selective axis inversion is being applied. This way we achieve the highest signal energy computable for a 3d-norm, which is illustrated by the purple graph.

A Butterworth band-pass is then applied to filter unwanted motion artifacts, other than walking. For the elderly, the following values were determined: lower_{cutoff} freq.: 0.25Hz; upper_{cutoff} freq.: 1.08Hz. While high-frequency artifacts may also be due to steps, they usually occur with healthy people who do not require a rollator. Where it is necessary to calculate the steps of a healthy user when objects are pushed or pulled, the upper_{cutoff} frequency can be opened to a slightly higher value, since more agile body movements and higher step impacts occur. A dynamic adjustment of the frequency filter is envisioned based on the currently dominant signal frequencies and signal magnitudes, which are both usually higher and also rather alternating at a healthy person.

The final number of steps taken is basically the overall number of local maxima, in a pre-defined range (up to 5Hz), multiplied by two, since the minimum values indicating a reverse arm swing are not counted. The counting of local maxima is performed by comparing a sample amplitude with its predecessor and successor value. In case the preceding and succeeding values are smaller, a maximum position is found. It must be noted, however, that this is only enabled due to the previously performed band-pass filtering, which erased multiple erroneous local maxima frequently occurring within a step.

5 EVALUATION

A controlled lab study, followed by a field study, was conducted to evaluate the performance of our step detection.

5.1 Lab Study (T1)

A study with eight participants (2 females), aged between 20-52yrs ($M=31.12$ yrs; $SD=9.89$ yrs) was conducted. The subjects had a body length of 1.69-1.96m ($M=1.835$ m; $SD=0.08$ m) and a calculated stride length between 0.7-0.81m ($M=0.76$ m; $SD=0.033$ m). All participants were wearing a SimValley AW 429 RX smartwatch on their left arm. We implemented our algorithm in a JAVA application running on that watch, while one of the experimenters manually kept track of the number of steps taken to gather ground truth data throughout the study. The subjects were asked to walk with a rollator the same pre-defined indoor route from the pilot study, which took 120-189 steps depending on the user's stride length and walking style. None of the subjects was advised on how to perform the walking to ensure unbiased and natural behavior.

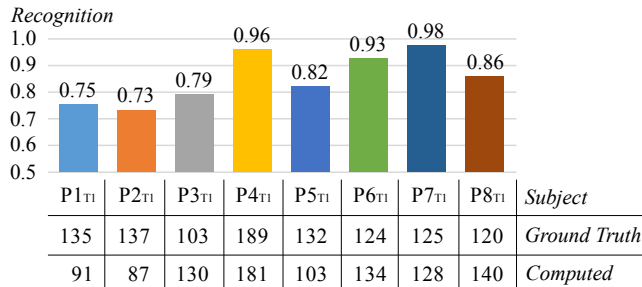


Figure 7. The table shows manually counted (ground truth) as well as computed steps and the resulting recall rate. The average recognition over all users is $M=85.22\%$, $SD=8.72\%$.

Based on Figure 7 we can see that our improved algorithm is indeed capable of recognizing the steps taken when using a rollator in a lab environment. The average recognition rate over all users of 85.22% is reasonably high.

5.2 Field Study (T2)

A field study was first conducted with five randomly selected elderly people (3 females) aged between 75-82yrs ($M=79.6$ yrs; $SD=2.45$ yrs). For this study, all subjects used their own individual rollator, while wearing the same smartwatch on their left arm. As before, the experimenter manually counted the steps taken by the subjects to gather ground truth data. The field study was conducted outdoors where the routes varied greatly. The results (see Figure 8) demonstrate a high ambiguity based on highly individual execution style, due to the differences in posture when holding the walking frame. The most obvious discrepancies occurred with participant $P3T2$, who used her arm to stretch out over the main frame, while bending over the rollator. This behavior resulted in poor recognition (~45%). The subject $P1T2$ repositioned the watch multiple times while walking, which led to substantial artifacts.

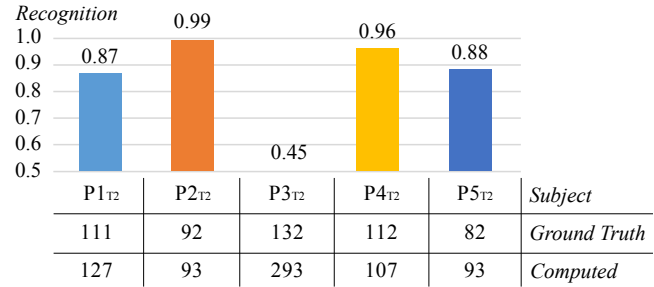


Figure 8. The table shows manually counted (ground truth) and computed steps. The recognition (recall rate) results in average over all users: $M=83.06\%$, $SD=21.85\%$.

The overall results, excluding $P3T2$, showed an average recognition rate of 92.56%. By including $P3T2$, a recognition of 83.06% could be reached, which is still reasonable.

5.3 Discussion

In the field, it was discovered that unforeseen events quickly occur, making average recognition rates above 90% difficult to achieve. A realistic scenario implies plenty of different artifacts, such as varying undergrounds, which can emit considerable artifacts (e.g., vibrations), individual walking styles (e.g., bending on the rollator while laying down the entire arm on the frame) etc. In our field study, elderly people who had different types of rollators and had different walking routes on different floor types, were selected. Given this, an average recognition of ~83.06% is thus promising. Comparing all three results ($M_{state-of-the-art}=10.35\%$; $M_{lab-study}=85.22\%$; $M_{field-study}=83.06\%$) show significant differences, as suggested by a weighted one-way ANOVA for independent samples ($F_{2,18}=70.25$; $p<.0001$). A Tukey HSD Test reveals our algorithm (in the lab as well as in the field) as significantly superior to current state-of-the-art algorithms, when comparing step-counting performance using a rollator. The field performance did not significantly differ from the findings in our lab ($p=.796$).

6 CONCLUSION

This research sought to improve accurate step-counting technology for the elderly, given its importance in quality health care. Current wrist-worn wearables are incapable of providing sufficient results when the elderly require a rollator. An improved step-counting algorithm was thus developed, significantly outperforming implementations deployed in current wearables. Preliminary observations, which includes tweaking the Butterworth filter's upper cutoff frequency, suggests that this newly developed algorithm is applicable in scenarios involving similar low-amplitude motions, such as when the user pushes a cart, a pram, a bike, including pulling or carrying a bag. Our algorithm is envisioned to operate independently, while using a dynamic adjustment of the frequency filter. Alternatively, the algorithm could also operate in a hybrid mode, parallel to an existing step detection algorithm, complementing the existing algorithm when it fails. Such developments will thus enable greater accuracy in step detection for low-amplitude motions in future wrist-worn wearables.

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